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MILITARY SPECIFICATION

# FLAGS AND PENNANTS

This specification is approved for use by all Departments and Agencies of the Department of Defense

## 1. SCOPE

1.1 Scope. This specification covers the general requirements for flags and pennants (other than U.S. National flags) used by the Department of Defense. The detailed requirements for a given flag will be covered by the drawing for that specific flag.

1.2 Classification. Flags and pennants shall be of the following materials and styles as specified (see 6.2):

MATERIALS
- Cotton Bunting
- Nylon, Lightweight Bunting
- Nylon, Heavyweight Bunting
- Nylon-Wool Bunting
- Acrylic Bunting
- Rayon Banner Cloth
- Nylon Cloth, Fluorescent

STYLES (DESIGN APPLICATION)
- Appliqued
- Hand Embroidered
- Machine Embroidered
- Screen Printed (not applicable to acrylic flags)
- Combination of above, as specified

1.2.1 Headings. The heading for flags and pennants shall be of the following types and classes as specified (see 6.2 and 6.5).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in preparing this document should be addressed to: Director, The Institute of Heraldry, US Army, Cameron Station, Alexandria, VA 22304-5050 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document.

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Type I - Duck

Class 1 - With distance line (except for acrylic flags)
Class 1A - With distance line (for acrylic flags)
Class 2 - With grommets
Class 3 - With distance line and thimbles
Class 4 - With grommets and leather tabs

Type II - Tube and tunnel type (with cotton sheeting lining and leather tab)

Type III - Hemmed
Class 1 - With tapes
Class 2 - With leather tabs

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

DDD-L-20 - Label, For Clothing, Equipage and Tentage (General Use)
NN-P-71 - Pallets, Material Handling, Wood Stringer Construction, 2-Way and 4-Way (Partial)
V-T-285 - Thread, Polyester
V-T-295 - Thread, Nylon
V-T-301 - Thread, Silk
UU-P 268 - Paper, Kraft, Wrapping
UU-P-553 - Paper, Wrapping, Tissue
CCC-C-419 - Cloth, Duck, Cotton, Unbleached, Plied-Yarns, Army and Numbered
CCC-C-430 - Cloth, Sheeting, Cotton and Polyester and Cotton
CCC-C-439 - Cloth, Bunting, Cotton, Mercerized
CCC-C-447 - Cloth, Percale, Cotton
CCC-C-476 - Cloth, Bunting, Nylon and Nylon and Wool
PPP-B-636 - Box, Shipping, Fiberboard
KK-L-2004 - Leather, Cattlehide, Deerskin and Horsehide, Chrome Tanned

MILITARY

MIL-H-226 - Halyard, Signal, Braided, Treated
MIL-C-606 - Cloth, Banner, Rayon
MIL-T-713 - Twine, Fibrous, Impregnated, Lacing and Tying
MIL-F-2310 - Fringe, Ornamental, Rayon
MIL-C-3636 - Cords and Tassels, Flag
MIL-T-5038 - Tape, Textile and Webbing, Textile, Reinforcing, Nylon
MIL-P-15011 - Pallets, Material Handling, Wood Post Construction, 4-Way Entry


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MIL-H-15021 - Hooks, Snap Bolt, Swivel-Eye and Rings.
MIL-G-16491 - Grommet, Metallic, General Specification For
MIL-R-17343 - Rope, Nylon
MIL-C-22775 - Cloth, Bunting, Acrylic
MIL-L-40051 - Leather, Cattlehide, For Glove Leather
MIL-T-43566 - Tape, Textile, Cotton or Polyester, General Purpose, Natural or in Colors

STANDARDS

FEDERAL

FED-STD-191 - Textile, Test Methods
FED-STD-751 - Stitches, Seams and Stitchings

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Standardization Document Order Desk, Building 4D, Section D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

COMMERCIAL ITEM DESCRIPTIONS

A-A-52094 Thread, Cotton

DRAWINGS

THE INSTITUTE OF HERALDRY (TIOH)

5-1-45 Series - Flag Headings

(Copies of specifications, standards, drawings and specification sheets required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

THE COLOR ASSOCIATION OF THE UNITED STATES

The Standard Color Reference of America, Tenth Edition.



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United States Army Color Card, The Official Colors for Branches, Agencies, etc.

DOD Standard Shades for Heraldic Yarns for Embroidery and Ribbons, 1974.

DOD Standard Color or Official Standardized Shades for Sewing Threads, 1968.

(Application for copies of the color cards should be addressed to the Color Association of the United States, 343 Lexington Avenue, New York, NY 10016-1187.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D-3951 - Standard Practice for Commercial Packaging

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other information services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard sample. Unless otherwise specified (see 6.2), samples shall be furnished by the contracting officer to be used for guidance, information and as a standard for design and workmanship. Variation from this specification may appear in the guide sample in which event the specification shall govern.

3.2 First article. Unless otherwise specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.7).

3.3 Material. Materials shall conform to the referenced specifications and the requirements specified herein.

3.3.1 Basic flag materials.

3.3.1.1 Cotton bunting. Cotton bunting shall conform to CCC-C-439.

3.3.1.2 Lightweight nylon bunting. Lightweight nylon bunting shall conform to Type I, Class A of CCC-C-476. At the option of the manufacturer, heavyweight nylon bunting conforming to material cited in 3.3.1.3 may be substituted for the lightweight nylon bunting except when the design is screen printed.

3.3.1.3 Heavyweight nylon bunting. Heavyweight nylon bunting shall conform to Type I, Class B of CCC-C-476 except that paragraph 3.6.1 shall be deleted and the following substituted: The finish of the nylon bunting shall be equal to or better than the standard sample.



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3.3.1.4 Nylon-wool bunting. Nylon-wool bunting shall conform to Type II of CCC-C-476. As an alternative, a 100 percent polyester fabric of plain weave may be used. The polyester fabric shall weight at least 4.5 ounces per square yard with a minimum of 50 yarns per inch in the warp and 32 yarns per inch in the filling.

3.3.1.5 Rayon. The rayon banner cloth shall be 50 denier conforming to MIL-C-606. As an alternative, a rayon fabric with the warp using two ply 60 denier (plus or minus 5) and single or two ply 50 denier filling may be used. The number of ends in the warp will be 230 (plus or minus 10). Other physical and test requirements specified in MIL-C-606 shall apply.

3.3.1.6 Fluorescent nylon cloth. Fluorescent nylon cloth shall be 200 denier nylon in both warp and filling, 3.0 ± 1/2 ounce per square yard, plain weave and heat treated. Fabric shall be fluorescent under ultraviolet light.

3.3.1.7 Cotton percale. Cotton percale shall conform to CCC-C-447.

3.3.1.8 Acrylic bunting. Acrylic bunting shall conform to MIL-C-22775.

3.3.2 Duck. The duck used for the heading shall be white or in the gray, shall weigh not less than 8.25 ounces per square yard, have 2 ply yarn in both warp and filling and shall have a breaking strength of not less than 105 pounds in the warp and 100 pounds in the filling direction when tested as specified for Type III, 8.25 ounce duck in CCC-C-419.

3.3.3 Sheeting.

3.3.3.1 Tube heading lining. Unless otherwise specified, the cotton sheeting used for lining the tube heading shall conform to Type I or II, Class 1, Style A or B of CCC-C-430.

3.3.3.2 Sheeting for insignia, stars or other symbols.

3.3.3.2.1 Cotton and mixed nylon-wool flags. Unless otherwise specified, insignia, stars or other symbols for cotton or mixed nylon-wool flags shall be made from cotton sheeting conforming to Type II, Class 1, Style A of CCC-C-430. The insignia, stars or other symbols may be sized or stiffened to aid in sewing provided the agent does not have a deleterious effect on appearance or serviceability.

3.3.3.2.2 Nylon flags. Unless otherwise specified, insignia, stars or other symbols for nylon flags shall be made from nylon conforming to Type I, Classes A or B of CCC-C-476. The insignia, stars or other symbols may be sized or stiffened to aid in sewing provided the agent does not have a deleterious effect on appearance, color or serviceability.

3.3.3.2.3 Rayon flags and acrylic flags and pennant. Unless otherwise specified, insignia, stars and other symbols for rayon flags and acrylic flags and pennants shall be made from the same material as the flag.

3.3.4 Thread.

3.3.4.1 Cotton. Cotton thread shall conform to A-A-52094. The types and sizes of thread shall be as specified in Table I.



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TABLE I. COTTON THREAD SIZES

| Flag or Flag Component | Type | Ticket No. | Ply | Finish |
|---|---|---|---|---|
| Cotton bunting flags | I | 50 | 3 | Soft |
| Rayon or Fluorescent Nylon flags | III | B | 3 | - |
| Distance lines | I or II | 16 | 4 | Soft or glazed |
| Cotton duck headings | I or II | 10 | 3 | Soft or glazed |
| Applique of cotton design components nylon-wool flag | I | 50 | 3 | Soft |

3.3.4.2 Nylon. Nylon thread shall conform to V-T-295. The thread for sewing nylon-wool flags and the hems and reinforcement patches of the nylon flags shall be machine twist or twisted bonded multicord, conforming to Type I or II, Class 1, Size B, 2 or 3 ply nylon thread. The thread used for sewing nylon flags, except for the hem and reinforcement patches, shall be Type I or II, Size A or B, 3 ply nylon thread.

3.3.4.3 Silk.

3.3.4.3.1 Hand embroidery. The silk thread for hand embroidery shall conform to Type IV of V-T-301 with the following exceptions to the requirements:

| Commercial Designation | Minimum Average Length Per Ounce (yard) | Minimum Average Break Strength (Pound) |
|---|---|---|
| A | 1000 | 3.0 |
| FF | 233 | 11.0 |

As an alternative, a ss mercerized and gassed cotton yarn may be used in lieu of the silk thread and may be substituted as follows:

| Silk Thread | Cotton Size |
|---|---|
| A Type | 20/2 |
| FF Type | 10/2 |

(See paragraph 3.3.4.3.2 for colorfastness requirements for above yarn.)



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3.3.4.3.2 Colorfastness for embroidery cotton yarn. Cotton yarn shall show good colorfastness to wet-dry cleaning, water, light and crocking.

3.3.4.4 Rayon. Rayon thread used in Bonnaz machine embroidery shall be 600 denier, continuous filament, regenerated cellulose (viscose or cuprammonium) rayon thread. The thread shall have not less than 460 yards per ounce and a breaking strength of not less than 2.3 pounds.

3.3.4.5 Polyester. Polyester thread for sewing acrylic flags and pennants shall conform to Type I, Class A, letter size B, 3 ply of V-T-285 and shall be white in color.

3.3.4.6 Color and colorfastness. Unless otherwise specified, the color of the thread shall match the color of the material on which it is used. Except for reinforcement patches, multicolored flags and pennants made of cloth other than acrylic bunting may be sewn with white thread or thread of the predominant color. Acrylic flags and pennants shall be sewn with white thread. The thread for duck headings shall be white or unbleached. When a fringe is specified, the color of the of the attachment threads for multicolored flags, visible on the reverse side of the flag shall match the predominant color of the base material. The colorfastness requirements of the thread used shall be equal to or better than the material on which it is used as described in 3.4.1.1.1.

3.3.5 Distance lines. The distance lines shall be 1/4 inch (0.64 cm) in diameter nylon rope conforming to MIL-R-17343 or 1/4 inch (0.64 cm) in diameter twisted or solid braided polypropolene rope having a minimum breaking strength of 1000 pounds as shown on the applicable drawing.

3.3.5.1 Twine. The twine used in wrapping the free end of the distance line shall be waxed twine conforming to Type N, Class 3 of MIL-T-713.

3.3.6 Tape. Tape shall be 1/2 inch (1.27 cm) nylon tape conforming to Type III of MIL-T-5038.

3.3.7 Snap hook and ring. The snap hooks and rings shall conform to MIL-H-15021. Unless otherwise specified on the applicable drawing, the snap hook shall conform to either Type I or Type II.

3.3.8 Grommets. Grommets shall conform to Type II, Trade Size 2 of MIL-G-16491/1 or Type III, Trade size No. 2 of MIL-G-16491(MS2030). Grommets shall have a bright finish.

3.3.9 Leather. Leather for the tabs shall conform to Type I, Class 2 of KK-L-2004, 3 to 4 ounces when measured on a Woburn Gauge or Type II of MIL-L-40051. When natural colored leather is used, the crocking requirements shall not apply.

3.3.10 Fringe. Except for automobile flags and unless otherwise specified, the fringe shall conform to Class 2 of MIL-F-2310. The fringe for automobile flags, when specified, shall conform to Class 1 of MIL-F-2310. The color of the fringe shall be as specified on the applicable drawing.

3.3.11 Webbing. When specified, the webbing used for reinforcement of Type I, Class 3 headings shall conform to Type I, Class 1 or 2 of MIL-T-43566 and shall be of the width specified on the applicable drawing.



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3.3.12 Cord and tassels. When specified (see 6.2), cord and tassels shall conform to MIL-C-3636 and the applicable drawing.

3.3.13 Thimbles. Thimbles shall be galvanized steel and of the type used for 1/4 inch (0.64 cm) in diameter wire rope. They shall be semicircular at the closed end and tapering at free ends.

3.4 Design and tolerance. The design and size of the flags and pennants shall be as specified on the applicable drawing. Unless otherwise specified on the drawing, a tolerance of plus or minus 5 percent shall be allowed for all dimensions exceeding 2 inches (2.54 cm) except that maximum and minimum measurement of dimensions on a single flag may not differ more than 2 percent of the required dimensions and both be within 5 percent tolerance. For dimensions 2 inches (2.54 cm) or under, a tolerance of plus or minus 1/8 inch (0.32 cm) shall be allowed.

3.4.1 Components of design. Components of the design shall be applied by one or a combination of the methods listed below and as specified on the applicable drawing. The face side of the flag shall be designated as the side visible when the heading is on the viewers left. Unless otherwise specified on the drawing, letters and numerals shall read from left to right on both sides of the flag.

(a) Applique. Appliqued designs shall be identically oriented in the location shown on the applicable drawing and shall appear proper on each side of the flag. Unless otherwise specified and except for acrylic flags or where components of design are printed on percale, the thread used for appliqueing shall match the color of the design component. Unless otherwise specified, components of design printed on percale shall be appliqued with thread matching the basic color of the percale. Components of design for acrylic flags and pennants shall be appliqued with white polyester thread specified in 3.3.4.5.

(b) Hand embroidery. Hand embroidered components of the design shall be embroidered in the location shown on the applicable drawing using thread specified in 3.3.4.3. Components of the design shall appear proper on the face side of the flag, be clearly distinguishable and show identical on the opposite side of the flag as if printed through the material.

(c) Machine embroidery. Machine embroidered components of the design shall be embroidered in the location shown on the applicable drawing using thread specified in 3.3.4.4. Components of the design shall appear proper on the face side of the flag, be clearly distinguishable, symmetrical and show identical on the opposite side of the flag as if printed through the material.

(d) Screen printing. Screen printed components of the design shall be printed in the location shown on the applicable drawing using screens made by the photographic method. Components of the design shall appear proper on the face side of the flag, be clearly distinguishable, symmetrical and show identical on the opposite side of the flag as if printed through the material. Acrylic flags shall not be screen printed.



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3.4.1.1 Color. Colors shall be as specified on applicable drawings and shall conform to approved shade standard samples furnished by the contracting officer or the shade standards cited in the latest edition of the Standard Color Reference of America (see 2.2). If cable numbers cited on applicable drawings do not match those cited in the Standard Color Reference of America, the outdated number will be converted to the current cable number. When cable number 70001 is cited, the number shall be converted to 80001. The colorant system for screen printed designs shall maintain the essential textile appearance of the base material. The dye shall not adversely effect the hand of the fabric. Coatings shall not be used unless approved by the contracting officer. When prints are required to register, the design shall be aligned when viewed by through lighting. When prints are applied to fabric for subsequent applique, some strike-through of the print is permissible provided it does not adversely affect the appearance of the finished flag or pennant. When penetrating dye is used, the color on the face shall match the color specified on the applicable drawing. A slight change of color will be allowed on the reverse side.

3.4.1.1.1 Colorfastness. The colorfastness of the base material for flags and pennants shall conform to the requirements of the applicable specification for the material. Colorants for screen printed designs shall provide colorfastness equal to or better than the dyed base fabric. The colorfastness to crocking for napthol colors or black areas may be fair. Rating on an adjective basis shall be performed only when a standard sample of the exact material is not provided.

3.5 Construction. Flags and pennants shall be constructed in accordance with the applicable drawing and the requirements specified herein.

3.5.1 Basic construction. All raw edges shall be hemmed. A selvage end may be used at the top or bottom edge except that flags and pennants manufactured from nylon, acrylic bunting or rayon banner cloth shall be hemmed on all edges. Hems shall be turned to the back side of the flags except hems on fringed flags shall be turned to the front side of the flag. The ends of all stitching shall be backstitched not less than one inch (2.54 cm) when not caught in a seam or other stitching. Except for flags and pennants of acrylic bunting, when the size of the pennant necessitates the piecing of the basic material, the seam, whenever possible, shall be in the center two-thirds of the material being pieced. When the size of an acrylic flag or pennant necessitates piecing the material, the base cloth may be used with the warp running in the vertical or hoist except the pennant code flags representing the letters Y, M and V and the numerals 4, 5 and 6 shall have material for the strip cut on a bias and set in on the straight of the material.

3.5.2 Headings. Unless otherwise shown on the applicable drawing, headings for flags and pennants shall be as specified herein.

3.5.2.1 Type I (Duck). Type I headings shall be fabricated from duck specified in 3.3.2 and the requirements specified herein.

3.5.2.1.1 Class 1 (Heading with distance line (except for acrylic flags). Type I, Class 1 headings shall be constructed in accordance with the details and dimensions shown on TIOH drawing 5-1-45A (figure 1) and shall meet the following requirements:

(a) The heading shall be made by cutting a piece of duck 6 inches (15.24 cm) wide and 2 inches (5.08 cm) longer than the hoist of the flag. The



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edges and ends of the duck strip shall be turned-in 1 inch (2.54 cm) and the duck doubled over to form a 2 inch (5.08 cm) wide heading.

(b) The hoist end of the flag shall be turned-under 3/4 inch (1.91 cm) and may be sewn with a single row of stitching. The folded end of the flag shall then be inserted between the edges of the duck for a distance of 1 inch (2.54 cm) so the ends of the flag shall not be exposed. With the turned-in sides and ends of the duck even, the hoist end of the flag and the open side of the heading shall be stitched with a minimum of three rows of equidistant stitching 1/4 (0.64 cm) to 3/8 inch (0.95 cm) gauge (1/8 inch (0.32 cm) stitching margin).

(c) The flag or pennant shall have a distance line specified in 3.3.5 running through the heading. The line at the top of the heading shall pass through a ring specified in 3.3.7, doubled back and superimposed upon itself and sewn with zigzag stitches through both thicknesses of the line for a minimum distance of 2 inches (5.08 cm). The line shall then be drawn into the heading until the ring is approximately flush with the outer edge of the heading. The top of the heading shall then be closed by two rows of stitches 1/4 (0.64 cm) to 1/2 inch (1.27 cm) gauge, either up to or through the halyard. The distance line shall be secured within the heading by zigzag stitches at the top, center and bottom for a minimum of 2 inches (5.08 cm) at each location. The lower end of the heading shall then be closed by two rows of stitching in the same manner as the top.

(d) The snap hook specified in 3.3.7 shall be attached to the distance line by passing the line through the swivel loop, doubled back and superimposed upon itself for a distance of not less than 2 inches (5.08 cm) and sewn by zigzag stitching for a minimum of 2 inches (5.08 cm). Zigzag stitching at the top and bottom of the line shall pass through both thicknesses of the distance line. Prior to zigzag stitching, the free end of the end of the distance line shall be wrapped for a distance of not less than 1/2 inch (1.27 cm) with not less than 8 turns using waxed twine specified in 3.3.5.1. The overall length of the extended portion of the distance line and snap hook, measured from the lower edge of the heading, shall be 9 inches (22.86 cm) for signal flags and pennants and 24 inches (61 cm) for all others.

(e) The circumference of distance lines for flags and pennants having an area of less than 16 square feet shall be 3/4 inch (1.91 cm). Unless otherwise specified on the applicable drawing, the circumference of distance lines for flags and pennants with an area of 16 square feet and larger shall be 1 inch (2.54 cm). The stitch type shall be as specified in Table II. At the option of the manufacturer, a Type I, Class 1A heading may be used as an alternate method of construction.

3.5.2.1.2 Class 1A (Heading with distance line for acrylic flags). Type I, Class 1A headings shall be constructed as specified in 3.5.2.1.1 with the following exceptions:

(a) The heading shall be constructed in accordance with the details and dimensions of TIOH drawing 5-1-45H (figure 2). The heading shall be stitched to the hoist end with four rows of stitching and a stitching



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margin of 1/8 inch (0.32 cm). The stitching gauge shall be 1/4 (0.64 cm) to 3/8 inch (0.95 cm).

(b) The distance line shall be fabricated from 1/4 inch (0.64 cm) diameter nylon rope specified in 3.3.5.

(c) The width of the zigzag stitching shall cover no less than one half the diameter of each strand of rope. The end of the zigzag stitching shall be firmly secured by backstitching or overstitching or by increasing the number of stitches per inch at the ends for a minimum of 1/4 inch (0.64 cm).

(d) The turned back end of the distance line shall be zigzag stitched from the snap hook to the free end of the distance line within the heading.

(e) The free end of the nylon rope shall be wrapped as specified in 3.5.2.1.1 or may be taped or fused at the ends to prevent raveling. Both cutting and fusing of the nylon rope may be accomplished in one operation if a hot wire or blade is used for the operation.

3.5.2.1.3 Class 2 (Heading with grommets). Type I, Class 2 headings shall be constructed in accordance with the details and dimensions shown on TIOH drawing 5-1-45B (figure 3) and shall meet the following requirements. Stitch type shall be as specified in Table II. The heading will be made with a piece of duck 6 inches wide and 5 inches (12.7 cm) longer than the flag hoist to which it is to be attached. Both side edges shall be turned-in 1 inch (2.54 cm) and both ends turned-in 2-1/2 inches (6.35 cm). The hoist end of the flag shall be turned under 3/4 inch (1.91 cm) and may be sewn with a single row of stitching. The folded ends of the inserted flag hoist shall not be exposed. The raw edges of the ends of the duck strip shall be closed with a single row of stitching using a 1/8 inch (0.32 cm) stitching margin or as an alternate, the rows of stitching may be placed after the duck is doubled over. The duck strip shall then be doubled over to form a finished heading 2 inches (5.08 cm) wide. With the sides and ends of the duck strip even, two rows of stitching (not less than 3/16 inch (0.48 cm) stitching gauge, 1/8 inch (0.32 cm) stitching margin) shall be sewn through all thicknesses. Single rows of stitching shall be placed along the centerfold edge and ends of the duck strip using a 1/8 (0.32 cm) to 1/4 inch (0.64 cm) stitching margin. The duck heading shall be finished by securely clinching two No. 2 grommets specified in 3.3.8 in positions shown on figure 3 for Type I, Class 2 headings.

3.5.2.1.4 Class 3 (Distance line with thimble). Type I, Class 3 headings as shown on TIOH drawing 5-1-45G (figure 4), shall be constructed in accordance with the dimensions specified for Class 1 headings in 3.5.2.1.1 and shown on TIOH drawing 5-1-45A (figure 1). The class 3 headings shall be fitted with galvanized steel thimbles specified in 3.3.13 so as to provide a loop or eye not less than 1-1/2 inches (3.81 cm) in length.

3.5.2.1.5 Class 4 (Heading with grommets and leather tabs). Type I, Class 4 headings shall be constructed in accordance with the details and dimensions shown on TIOH drawing 5-1-45D (Figure 5) from duck specified in 3.3.2 and shall meet the following requirements:



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(a) The heading shall be made by cutting a piece of duck 8 inches (20.32 cm) wide and 5 inches (12.7 cm) longer than the hoist of the flag. Stitch type shall be as specified in Table II.

(b) The hoist end of the flag shall be positioned 1 inch (2.54 cm) in from the side edge of the duck, 2-1/2 inches (6.35 cm) from each end and stitched with one row of stitching. The side edge of the duck attached to the flag shall be turned-in 1 inch so that a 1-inch (2.54 cm) strip of flag material the same length as the hoist shall be folded inside the turned-in duck fold as shown on the applicable drawing. The opposite side edge of the duck shall also be turned-in 1 inch (2.54 cm). The ends of the duck shall be turned-in 2-1/2 inches (6.35 cm). The raw edges of the top and bottom duck ends shall be stitched closed with a single row of stitching with a 1/8 inch (0.32 cm) stitching margin.

(c) A tab, made of leather specified in 3.3.9, 3/4 inch ± 1/16 inch (1.91 cm ± 0.16 cm) wide and not less than 2 inches (5.08 cm) long, having a slit and eyelet 3/4 inch ± 1/16 inch (1.91 cm ± 0.16 cm) long, for fastening the tab over a knob or button of the flagstaff shall be centered and placed flush with each end of the heading as shown on TIOH drawing 5-1-45D (figure 5). The tabs shall be secured through the two thicknesses of duck by box stitching as shown on the applicable drawing. Stitching shall be securely backstitched.

(d) Four No. 2 grommets specified in 3.3.8 shall be clinched into position as indicated on the applicable drawing. The duck heading shall be doubled over to form a heading width of at least 2-3/4 inches (6.99 cm). Position the sides and ends of duck evenly, with the grommet holes aligned and sew two rows of stitching not less than 1/16 inch (0.16 cm) gauge, 1/8 inch (0.32 cm) margin through all thicknesses.

3.5.2.2 Type II (Tube and tunnel type heading with cotton sheeting lining and leather tab). Type II heading shall be constructed in accordance with the details and dimensions shown on TIOH drawing 5-1-45E (Figure 6) from cotton sheeting specified in 3.3.3.1 and shall meet the following requirements:

(a) The heading shall be made by cutting a piece of sheeting 6 inches (15.24 cm) wide and 5 inches (12.7 cm) longer than the hoist of the flag; guidon sheeting shall be cut down proportionately in the width so a 2-1/4 inch (5.70 cm) heading is formed. Each end shall be folded under 2-1/2 inches (6.35 cm).

(b) The sheeting shall be fitted with a tab made of leather specified in 3.3.9 at each end. Each tab shall be 3/4 inch ± 1/16 inch (1.91 cm ± 0.16 cm) wide and not less than 2 inches (5.08 cm) long. Each tab shall have a slit and eyelet 3/4 inch ± 1/16 inch (1.91 cm ± 0.16 cm) long for fastening over a knob or a bottom on the flagstaff. The slit and eyelet shall be centered on the width of the tab shown on the applicable drawing. The tab shall be placed within the heading and centered opposite the stitching of the hem. The tab shall be secured through the two thicknesses of lining by box stitching as shown on the applicable drawing. As an alternate, the leather tabs may be secured by stitching through both lining and heading material. The stitching



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shall be securely backstitched. The contractor's label specified in 3.8 shall be affixed at the same time the tabs are placed on the sheeting.

(c) The sheeting strip shall be placed on the back of the flag at the hoist end with the folded ends of the strip flush with the hemmed or salvage edges of the flag at the top and attached thereto by a single row of stitching across each folded end of the strip. The raw edge of the stitching shall be placed 1/2 inch (1.17 cm) from the raw edge of the flag. The tube or tunnel shall then be formed by hemming the hoist end of the flag with the sheeting strip inside. Sheeting may also be seamed separately with a 1/2 inch (1.27 cm) seam to form a tunnel. Turn under hoist edge of flag 1/2 inch (1.27 cm) and insert under the folded side edge of sheeting strip. Turn under and stitch with a single or double row of stitching (3/16 inch (0.48 cm) margin), which shall pass through both the turned edge of the flag and both edges of the unseamed or seamed lining strip. The flat width of finished hem shall be not less than 2-3/4 inches (6.99 cm) (except for guidons which shall not be less than 2-1/4 inches (5.70 cm).

3.5.2.3 Type III (Flags without duck or cotton sheeting headings). Type III headings shall have the hoist end of the flag hemmed and shall be constructed in accordance with the requirements specified herein.

3.5.2.3.1 Class 1 (with tape). Type III, Class 1 heading shall be constructed in accordance with the details and dimensions shown on TIOH drawing 5-1-45F (figure 7). Stitch type shall be as specified in two rows of stitching. Nylon tape specified in 3.3.6 shall be cut 12 inches long (30.48 cm), doubled end-to-end so as to form tie-tapes and attached to the hem with two rows of stitching. One tape shall be attached 1 inch (2.54 cm) from the top of the hem and the other 1 inch (2.54 cm) from the bottom. The tapes shall overlap the full width of the hem and be arranged so as to facilitate the tying of the flag to a flagstaff.

3.5.2.3.2 Class 2 (with leather tabs). Type III, Class 2 headings (automobile and other flags when required) shall be constructed in accordance with the details and dimensions shown on TIOH drawing 5-1-45F (figure 7). Two rectangular leather tabs 1-3/8 inch (3.49 cm) long and 5/8 inch (1.59 cm) wide, shall be located and secured within the hem as shown on the applicable drawing. After insertion of the tabs, the hoist end shall be hemmed.

3.5.3 Stitching of fringe. When fringe is specified for flags, a continuous length of the fringe conforming to 3.3.10 shall be stitched to the front side of the flag at the top, fly and bottom edge of the flag and completely around each end of the heading. The fringe shall be neatly turned under and caught in the heading stitching. Selvage edges at top and bottom shall be hemmed with Type EFa-1 stitching, cut edges with EFb-1 stitching. The stitching used to attach the fringe may be incorporated in the formation of the hem. As an alternative, edges may be hemmed with an EFa-1 stitching and turned over prior to attachment of fringe. The fringe shall be superimposed on the hem and evenly attached around the flag with a single row of stitching through the center of the fringe using seam type LSa-1, except that automobile flags will have fringe attached with three rows of stitching, seam type LSa-1.



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3.6 Stitches, seams and stitching. Unless otherwise specified, stitches, seams and stitching shall conform to FED-STD-751 and as shown in Table II.

TABLE II. STITCHES, SEAMS AND STITCHING

| Section of flag or pennant | Type of seam or stitching | Minimum hem width (Inch) | Stitch Type | Stitches per inch |
|---|---|---|---|---|
| Hem (Top and Bottom): | | | | |
| Flags with fringe.... | EFa-1 or EFb-1 | 3/32 - 3/16 | 301 | 12 |
| Flags, nylon..... | EFa-1 or EFb-1 | 3/16 | 301 | 12 |
| All other flags 1/... | EFb-1 or EFb-2 | 3/8 | 301 | 12 |
| Tail or fly end: | | | | |
| Flags with fringe.... | EFb-1 | 3/32 - 3/16 | 301 | 12 |
| Semaphore, wigway and flags with Type II headings..... | EFb-2 | 1/2 | 301 | 12 |
| All other flags..... | EFb-3 or EFb-4 | 3/4 | 301 | 12 |
| Reinforcement patches | LSd-1 or LSd-2 | - | 301 | 12 |
| Seams: 5/ | | | | |
| Cross and fly..... | LSc-2 | - | 301 | 12 |
| Bias..... | LSr-2 or LSc-2 | - | 301 | 12 |
| Appliqued design: | | | | |
| Cotton 3/..... | - | - | 304 | 24 |
| Nylon, Nylon-Wool, Acrylic 4/..... | - | - | 304 | 28 to 32 |
| Rayon 2/..... | - | - | 304 | 58 |
| Headings: | | | | |
| Type I (Ducks): | | (Heading width) | | |
| Class 1..... | 3.5.2.1.1 | 2 | 301 | 6 |
| Class 1A..... | 3.5.2.1.2 | 2 | 301 | 6 |
| Class 2..... | 3.5.2.1.3 | 2 | 301 | 6 |
| Class 3..... | 3.5.2.1.4 | 2 | 301 | 6 |
| Class 4..... | 3.5.2.1.5 | 2-3/4 | 301 | 6 |
| Type II (Tunnel type) for all flags except guidons..... | 3.5.2.2 | 2-3/4 | 301 | 12 |
| Guidons..... | 3.5.2.2 | 2-1/4 | 301 | 12 |
| Type III (hemmed): | | | | |
| Class 1..... | 3.5.2.3.1 | 1/2 | 301 | 12 |
| Class 2..... | 3.5.2.3.2 | 1-1/4 ± 1/4 | 301 | 12 |
| Distance lines..... | - | - | 304, 308 or hand stitched | 6 |
| Fringe..... | LSa-1 | - | 301 | 12 |



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1/ Seam types of the same letter designation (e.g., EFb, LSd) are the same seams and differ only in the number of rows of stitching as designated by the number following the dash (e.g., EFb-1, EFb-2 or LSd-1, LSd-2).
2/ Throw of stitch 1/16 inch (0.16 cm) (unless otherwise specified on drawing).
3/ Throw of stitch 1/8 inch (0.32 cm) (unless otherwise specified on drawing). A 1/16 inch (0.16 cm) throw is permitted if stitching area is restricted in size.
4/ Throw of stitch 1/6 (0.16 cm) to 1/8 inch (0.32 cm) (unless otherwise specified on drawing).
5/ As an alternative, seams of signal flags and signal pennants may be stitched with a combination of SSa-2 seam and a LSa-2 seam. After completion of the SSa-2 seam with stitch type 515, the edge shall be smoothed and the seam type LSa-2, stitch type 301, shall be applied within 1/8 inch (0.32 cm) from the edge with a 1/4 inch (0.64 cm) gauge. The throw of the SSa-1 seam shall be 3/8 inch (0.95 cm). The SSa-2 and LSa-2 shall have a minimum of 12 stitches per inch.

3.7 Reinforcement patches. When required, all patches shall be located on the reverse side of the flag or pennant and shall be fabricated of the same material in the same color as that of the section to be reinforced. All exposed edges shall be turned under a minimum of 1/4 inch (0.64 cm) and securely stitched. Reinforcement patches are not required on the following:

(a) Flags with Type II and III headings.

(b) Flags with a hoist of 24 inches (60.96 cm) or less.

(c) Pennants made from acrylic bunting.

(d) International signal flags made from acrylic bunting except for sizes 3, 4 and 6.

3.7.1 Cotton and nylon-wool bunting flags. Reinforcement patches on cotton and nylon-wool bunting flags with Type I headings shall be sewn to each corner at the hoist end of the flag and shall extend within the heading the same distance as the flag base. The size of the reinforcement patches shall be as shown on the applicable drawing or at the option of the manufacturer, the patches may be 4 inches (10.16 cm) square for flags with hoist lengths of 2 (60.96 cm) to 3 feet (91.44 cm) or 6 inches (15.24 cm) square for flags with hoist lengths over 3 feet (91.44 cm). The size of the patch shall be as near the required size as possible.

3.7.2 Nylon and acrylic bunting flags. Reinforcement patches on nylon and acrylic bunting flags with Type I headings shall be sewn in each corner of the fly end of the flag. The patches shall be in the shape of right angle triangles with the apex (90 degree angle) located at each corner of the fly end of the flag. The size of the reinforcement patches shall be as shown on figure 9. Sewing of the patches to the fly end of the flag shall be accomplished by folding the top or bottom edge (as applicable) of the flag base 3/8 inch (0.95 cm) over the patch and then refolding both plies of fabric over 3/8 inch (0.95 cm) (see figure 8) and stitching 1/16 inch (0.16 cm) from the inner and outer edges of the fold to form the top and bottom hems. The fly end hem shall be formed by folding the edge of the base material 3/4 inch (1.91 cm) over the patch and then refolding both plies of fabric over 3/4 inch (1.91 cm)(see figure 8) and stitching the hem with three evenly spaced rows of stitches. The outer rows of stitching shall be not more than 1/16 inch (0.16 cm) from the outer edges of the hem.



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3.8 Marking. Flags and pennants with Type I headings shall be marked on the heading in accordance with a Type IV, Class 9 label of DDD-L-20 with letters appropriate to the size of the heading but not more than 3/4 inch (1.91 cm) high. The lettering shall run along the length of the heading, beginning immediately below the upper grommet on Class 2 headings and in a corresponding position on all other Type I headings. Marking shall contain information required for Class 5 labels and shall be equal to Class 8 labels in fastness of printing. For Type II headings, a label conforming to Type I, Class 5 of DDD-L-20 shall be affixed adjacent to the top leather strap.

3.9 Workmanship. Flags and pennants shall be clean, well made and shall meet the acceptable quality levels established by this specification. All stitching which is not caught up by a seam or stitching shall be backstitched a minimum of one inch (2.54 cm). Thread tension shall be maintained throughout and all thread ends shall be trimmed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.5.

4.4 First article inspection. Inspection and testing of the first article shall be made on a completely fabricated flag or pennant for all provisions of this specification applicable to the end product examination and tests (see 6.7).

4.5 Quality conformance inspection. Inspection shall be in accordance with the



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provisions set forth in MIL-STD-105, except where otherwise indicated.

4.5.1 Testing of components. In addition to the quality assurance provisions of the subsidiary specifications and standards, inspection shall be performed on components and materials listed in table III for test characteristics shown therein. The inspection level shall be S-1 and the acceptable quality level (AQL) shall be 6.5 defects per hundred units. The sample unit for the inspection shall be one cone or spool.

TABLE III. COMPONENT TESTING

| Component and lot expressed in terms of | Characteristic | Reqm't Ref | Test Method 1/ | Reqm't Individual Unit | Appl to lot average | #deter per sample unit |
|---|---|---|---|---|---|---|
| Silk embroidery thread | Breaking strength | 3.3.4.3.1 | 4104 | - | X | - |
| | Length weight | | 4010 | - | X | - |
| Cotton yarn | Color-fastness to Wet-dry cleaning | 3.3.4.3.2 | 5622 | - | - | - |
| | Water | | 5630 | - | - | - |
| | Light | | 5660 | - | - | - |
| | Crocking | | 5661 | - | - | - |
| Bonnaz rayon embroidery thread | Ply 2/ | 3.3.4.4 | Visual | X | - | 1 |
| | Breaking strength | | 4102 | - | X | - |
| | Length-weight | | 4010 | - | X | - |
| | Material 2/ identification | | Std comm'l | X | - | 1 |

1/ Test will be in accordance with FED-STD-191.
2/ Results reported as pass or fail.

4.5.2 In-process inspection. Inspection shall be performed at any point or during any phase of the manufacturing process to determine whether operation or assemblies are accomplished as specified. The government reserves the right to exclude from consideration for acceptance any material for which in-process inspection has indicated non-conformance. In-process inspection shall be conducted to see that accomplishment of the following is in accordance with the specification requirements.



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| Requirement, operation or assembly | Requirement paragraph |
|---|---|
| Heading construction | 3.5.2 thru 3.5.2.3.2 as applicable |
| Machine thread shall be checked at the machine to determine that threads, as specified, are used | 3.3.4 thru 3.3.4.6 as applicable |
| Turn-in and turn-under of seams, hems, reinforcement patches and headings. | 3.5.2 thru 3.5.2.3.2 and 3.7 thru 3.7.2, as applicable |

4.5.3 Intermediate testing. Testing of the partially fabricated flag or pennant shall be performed in accordance with Table IV for the characteristics shown therein. The sample unit shall be one screen printed design for appliqued flags or one screen printed flag base as applicable, prior to sewing into the completed flag. The inspection level shall be S-1 and the acceptable quality level (AQL) for each characteristic shall be 6.5 defects per hundred units. All requirements shall be applicable to the sample unit.

TABLE IV. TESTING OF THE INTERMEDIATE PRODUCT

| Screen printed designs for: | Colorfastness to: | Requirement Paragraph | Test method 3/ |
|---|---|---|---|
| Cotton, nylon and nylon-wool flags | Light | 3.4.1.1.1 | 5660 1/ |
| | Crocking | 3.4.1.1.1 | 5651 |
| | Water | 3.4.1.1.1 | 5630 2/ |
| Rayon | Light | 3.4.1.1.1 | 5660 1/ |
| | Crocking | 3.4.1.1.1 | 5651 |
| | Water | 3.4.1.1.1 | 5630 2/ |
| | Wet-dry Cleaning | 3.4.1.1.1 | 5622 |

1/ In the event of a dispute resulting from test method 5660, or as a result of suspected anomalous behavior of certain dye types or formulations, the contracting officer may authorize the exposure to natural light in accordance with method 5662.
2/ Method 5630 shall be used except the test temperature shall be 23.9 degrees to 29.4 degrees C (75 degrees to 85 degrees F).
3/ Test method in accordance with FED-STD-191.

4.5.4 Inspection of the end item.

4.5.4.1 Visual examination of flags and pennants. Examination shall be made at a distance of approximately 18 (45.72 cm) to 40 inches (101.6 cm) with good lumination. The defects found during examination shall be classified in accordance



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with 4.5.4.1.1 and 4.5.4.1.2. The lots shall be inspected in accordance with 4.5.4.1.3. The unit of product for these examinations shall be one completely fabricated flag or pennant as applicable.

4.5.4.1.1 Examination of flags or pennants for defects in material, design, construction, workmanship and marking. Defects designated by an asterisk (*) shall be classified as major when seriously affecting serviceability or appearance and minor when affecting serviceability or appearance, but not seriously.

Table V. DEFECTS

| Examine | Defect | Classification | | |
|---|---|---|---|---|
| | | Major | (*) | Minor |
| Quality of: | | | | |
| Type | Any component not fabricated of specified material | - | * | - |
| Color | Basic color other than specified | x | - | - |
| | Off shade | - | * | - |
| Material | Hole, cut, tear, smash or weak place | - | * | - |
| | Spots, stains, streaks or shade bars (considering the use position) | - | * | - |
| | Two or more continuous broken or missing ends or picks in cotton, acrylic, nylon and nylon-wool cloth | x | - | - |
| | Two or more continuous broken, missing ends or picks in rayon cloth | x | - | - |
| | Double pick, coarse pick or mispick clearly noticeable | - | * | - |
| | Slub, slough-off, float or skips | - | * | - |
| Quality of distance line: | | | | |
| Type | Not fabricated of specified material | - | - | x |
| Braiding | Uneven or loose resulting in weak place | x | - | - |
| | Any cut, chafing, fraying, or abrasion | - | * | - |
| Quality of metal components: | | | | |
| Snap hook & ring | Not fabricated of specified material | - | - | x |
| | Snap hook does not work freely; binds or catches when being opened or closed | - | * | - |
| | Broken, cracked or malformed | - | * | - |
| Grommets | Broken, cracked or malformed | - | * | - |
| | Contains rough or sharp edges | - | - | x |
| Quality of leather components | Not fabricated of specified leather | - | - | x |
| | Hole (other than required), cut, tear, or weak place | x | - | - |
| | Broken, cracked or malformed | - | * | - |



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Table V. DEFECTS (Continued)

| Examine | Defect | Classification | | |
|---|---|---|---|---|
| | | Major | (*) | Minor |
| Quality of fringe, cord and tassels | Material, type, construction, color, or color combination, not as specified..... | - | * | - |
| | Frayed edges..... | - | * | - |
| | Spotted, stained or soiled..... | - | * | - |
| | Cut or tear..... | - | * | - |
| | Wood mold not completely covered with braiding..... | - | - | x |
| | Wood mold loosely or unevenly covered..... | - | - | x |
| Screen Printing | Any part of design omitted..... | - | * | - |
| | Coating used in lieu of dye..... | x | - | - |
| | Poor registration, over printing or feathering..... | - | * | - |
| | Wrong color, or arrangement of colors..... | x | - | - |
| | Dye used in screen printing adversely affects hand of fabric..... | - | * | - |
| | Dye bronzed or mottled to extent of affecting appearance at 6 feet (182.88 cm).. | - | * | - |
| | Poor penetration of dye from front to back... | - | * | - |
| Construction and workmanship (General) | Operation omitted or not accomplished as specified..... | - | * | - |
| | Flag pieced where piecing is prohibited..... | x | - | - |
| | Component is not the required type, class or size..... | - | * | - |
| | Warp threads of the basic material of the flag in other than the fly direction of the flag, except as permitted in 3.5.1..... | - | * | - |
| | Any component missing..... | x | - | - |
| Stitches, seams and stitching | Wrong type of seam, stitches or stitching.... | - | * | - |
| | Wrong color of thread..... | - | * | - |
| | Row of stitches partially or completely missing..... | - | * | - |
| | Wrong stitching gauge or margin..... | - | * | - |
| | Seams improperly formed, material caught in unrelated stitching operation..... | - | * | - |
| | Ends of stitching not backstitched one inch (2.54 cm) (when not caught in other seams or stitching)..... | - | * | - |
| | Run-offs resulting in open seam..... | - | * | - |
| | Needle chews..... | - | * | - |
| | Any raw edges exceeding 1/2 inch (1.27 cm) in length..... | - | - | x |
| | Loose stitch tension resulting in a loosely secured seam or thread lying on surface of seam..... | x | - | - |



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Table V. DEFECTS (Continued)

| Examine | Defect | Classification Major | (*) | Minor |
|---|---|---|---|---|
| | Tight stitch tension (stitches break when normal pull is exerted in lengthwise direction of seam or stitching..... | x | - | - |
| | Stitches per inch for seam, hems and reinforcement patches-four or more stitches less than specified..... | x | - | - |
| | --one to three stitches less than specified. | - | - | x |
| | Stitches per inch for duck headings: | | | |
| | --two or more stitches less than specified.. | x | - | - |
| | --one stitch less than specified..... | - | - | x |
| | Stitches per inch for appliqueing: | | | |
| | --five or more stitches less than specified. | x | - | - |
| | --one to four stitches less than specified.. | - | - | x |
| | NOTE: Variation in the number of stitches per inch caused by the operator speeding up the machine or pulling fabric, such as sewing over heavy places, or in turning corners shall be classified as follows: Majors to minors, minors to no defect. | | | |
| | Seam twisted, pleated or puckered..... | - | * | - |
| | More than two continuous skipped stitches.... | - | * | - |
| | Broken stitches in heading seam causing an unstitched length of one inch or more..... | x | - | - |
| | Broken stitches in heading seam causing an unstitched length of less than 1 inch (2.54 cm)..... | - | - | x |
| | Broken stitches in any other seam causing an unstitched length of less than 1/2 inch (1.27 cm)..... | - | - | x |
| Embroidery Design (insignia,emblem, stars, lettering, etc.) | Broken stitches..... | - | * | - |
| | Not applied as specified..... | x | - | - |
| | Design or parts of design reversed or inverted..... | x | - | - |
| | Design misplaced or in wrong location..... | - | * | - |
| | Lettering does not read from right to left on both sides (unless otherwise specified).. | x | - | - |
| | Design not oriented on each side as specified | x | - | - |
| | Pattern marks used for location of design visible on finished product..... | - | * | - |
| | Design or insignia not neatly trimmed ..... | - | * | - |
| | Evidence that sizing or stiffening agent used as an aid in sewing has deleterious effect..... | - | * | - |
| Hems | Nylon or acrylic flags not hemmed on all edges..... | x | - | - |



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Table V. DEFECTS (Continued)

| Examine | Defect | Classification | | |
|---|---|---|---|---|
| | | Major | (*) | Minor |
| | Hem uneven | - | * | - |
| | Hem on wrong side of flag | - | - | x |
| | Hem puckering | - | * | - |
| Reinforcement Patches | Positioned on face side of flag | - | - | x |
| | Portion of patch at heading not inserted within the headings as specified | - | - | x |
| | Patch not the same material or color as section reinforced | - | * | - |
| | Size or shape of patch not as specified | - | * | - |
| Fringe | Not completely attached as specified | x | - | - |
| | Ends of fringe not caught in heading stitching | - | - | x |
| | Not one continuous length | - | - | x |
| | Not stitched to the front side of flag | - | - | x |
| | Not evenly attached around the flag | - | - | x |
| | Stitching of fringe not through center portion of the fringe heading | - | * | - |
| | One half of fringe not visible on reverse side of flag | - | * | - |
| Heading - Type I, Class 1 or or 1A or 3, as applicable | Lower end of line not passed through snap hook and sewed to itself | x | - | - |
| | Length of zigzag stitching not as specified | - | - | x |
| | Zigzag stitching not sewn completely through all thicknesses of heading and line at any of the required locations | - | * | - |
| | Lower edge of ring more than 1/2 inch (1.27 cm) above or within the upper end of heading | - | - | x |
| | Ends of heading not closed by two rows of stitching up to distance line | - | * | - |
| | Free end of distance line not wrapped, fused or taped as specified | - | * | - |
| | Sides and ends of duck headings out of alignment | - | * | - |
| Heading - Type I, Class 2 | Grommet clinched excessively tight, cutting fabric | x | - | - |
| | Grommet improperly set, e.g., clinched too loosely allowing grommet to rotate | - | - | x |
| | Sides and ends of duck out of alignment | - | * | - |
| Heading - Type I, Class 4 | Grommets clinched excessively tight, cutting fabric | x | - | - |
| | Grommets improperly located | - | * | - |



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Table V. DEFECTS (Continued)

| Examine | Defect | Classification Major | (*) | Minor |
|---|---|---|---|---|
| Type II | Sides and ends of duck heading not properly aligned | - | * | - |
| | Grommet improperly set, e.g., clinched too loosely allowing grommet to rotate | - | - | x |
| | Leather tab not positioned and attached as shown on drawing | - | - | x |
| | Sheeting all or partially missing | x | - | - |
| | Heading improperly formed, e.g., both edges of the sheeting or hoist end of flag not caught entire length by heading seam | - | - | x |
| | Leather tabs not positioned and attached as shown on drawing | - | - | x |
| | Heading seam not formed on back of flag | - | - | x |
| Heading - Type III, Class 1 | Tie tape missing | x | - | - |
| | Tie tape not positioned or attached as specified | - | - | x |
| Heading Type III, Class 2 | Hemmed heading not as shown on drawing | - | * | - |
| | Leather tab not positioned and attached as shown on drawing | - | * | - |
| Marking | Missing, incorrect, illegible, text incomplete, marking in wrong location, size of lettering not as specified or color marking not as specified | - | - | x |
| | Marking not permanent ink when required | - | - | x |
| Label | Text incomplete or incorrect, not attached as specified, not in location specified or not in position specified | - | - | x |
| | Marking not permanent ink when required | - | - | x |

4.5.4.1.2 Examination of flag or pennant for defects in dimensions.



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Table VI. DIMENSION DEFECTS

| Examine | Defect | Classification Major | Classification Minor |
|---|---|---|---|
| Overall length (minus heading): Overall width of flag or pennant | Not within the specified tolerance (± 5 % or ± 5/8 inch (1.59 cm) per foot..... | x | - |
| Width of heading: | Varies from specified finished width by: -1/2 inch (1.27 cm) or more from specified width..... | x | - |
| | -more than 1/4 inch (0.64 cm) but less than 1/2 inch (1.27 cm) specified width..... | - | x |
| Design (emblems, stars, insignia, | Dimensions of design or any part thereof out of balance or not in symmetrical alignment (check by comparing approved samples or drawing): -seriously affecting serviceability or appearance..... | x | - |
| | -affecting serviceability or appearance but not seriously..... | - | x |

4.5.4.1.3 Inspection levels and acceptable quality levels (AQLs).

The inspection levels and acceptable quality levels expressed in defects per hundred units shall be as follows:

| | Inspection level | AQL Major | AQL Total |
|---|---|---|---|
| For defects applicable to 4.5.4.1.1 | I | 4.0 | 10.0 |
| For defects applicable to 4.5.4.1.2 | S-3 | 6.5 | 15.0 |

4.5.5 Examination of count of flags and pennants in intermediate containers. Flags and pennants packaged for shipment shall be examined to determine conformance with package markings and specified quantity. The sample unit for this examination shall be one box (interior package). Any box containing less than the specified or marked quantity of flags and pennants shall be classified as a defect. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

| EXAMINE | DEFECT |
|---|---|
| Markings, exterior and interior | Omitted, incorrect, illegible or improper size, location sequence or method of application. |
| Materials | Any component missing. Any component damaged affecting serviceability. |
| Workmanship | Inadequate application of components, such as incomplete closure of case liners, container flaps, loose or inadequate strapping. |



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| | |
|---|---|
| | Bulging or distortion of containers. |
| Weight | Net weight exceeds the requirements. |

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Flags and pennants made of nylon-wool shall be thoroughly dusted over the entire area with flake napthalene before being packaged (see 6.3). Flags with distance lines shall have the exposed portion of all attached hardware wrapped or covered in polyethylene film 0.002 inch thick (minimum).

5.1.1.1 Unit packaging.

5.1.1.1.1 Flags and pennants of nylon, nylon-wool, acrylic and cotton bunting. Each flag or pennant, preserved as specified in 5.1.1, shall be compactly folded or folded and rolled and then placed in a snug-fitting transparent, end-opening, polyethylene bag. The polyethylene film shall be 0.003 inch thick (minimum). All seams and closures shall be made by heat sealing. Prior to or during the heat sealing operation, excess air within the bag shall be expelled. Closure shall be in accordance with the appendix of the applicable specification.

5.1.1.1.2 Flags and pennants of rayon banner cloth. Each flag or pennant shall be laid out flat on two thicknesses of tissue paper conforming to Type I, Grade A, Class 1 of UU-P-553. Tissue paper shall overlap all edges of the flag or pennant a minimum of two inches (5.08 cm). Each flag or pennant combined with overlapping tissue paper shall be rolled open width on a chipboard tube. The tissue overlapping the tube ends shall be tucked inside the tube ends. Each flag or pennant shall be tightly wrapped with 50-pound minimum basis weight kraft paper conforming to Grade B of UU-P-268. The closure shall be sealed with 2-inch (5.08 cm) minimum width gummed paper tape. Each wrapped flag or pennant shall then be individually packaged in a snug-fitting box conforming to PPP-B-636 Class Domestic, Style optional. Closure shall be in accordance with the appendix of the box specification.

5.1.1.2 Intermediate packing. Unless otherwise specified, flags or pennants of one material, type, style and class only, units packaged as specified in 5.1.1.1 shall be neatly stacked and packaged in multiples of 12 within a snug-fitting intermediate fiberboard container conforming to Class Domestic, Style RCS-L, Grade 275 of PPP-B-636. The net weight of contents in each container shall not exceed 65 pounds. Closure shall be in accordance with the appendix of the applicable specification.

5.1.2 Commercial. Assembled flags and pennants shall be preserved and packaged in accordance with the applicable requirements of ASTM-D-3951.

5.2 Packing. Packing shall be level A, B or commercial as specified (see 6.2).

5.2.1 Level A. Flags and pennants packaged as specified in 5.1 shall be packed in a snug-fitting shipping container conforming to W5c, W5s, V3c or V3s of PPP-B-636. Closure shall be in accordance with the applicable container specification.



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5.2.2 Level B. Flags and pennants packaged as specified in 5.1.1.2 shall be acceptable as level B packing.

5.2.3 Commercial. Assembled flags and pennants preserved as specified in 5.1, shall be packed in accordance with the applicable requirements of ASTM-D-3951.

5.3 Palletization. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be palletized on a 4-way entry pallet in accordance with load type 1A of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C, K and L or O or P. Pallet pattern shall be in accordance with the appendix of MIL-STD-147. The pallet shall be 4-way, type I, Class 1, Style 1, Size A, Wood Group I, II, III or IV of MIL-P-15011, or, 4-way, Type IV, V or VIII, Class 1, Style A, Size 2, Wood Group I, II, III or IV, Grade A of NN-P-71. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall be approved by the contracting officer.

5.4 Marking. In addition to any special markings required by the contract or purchase order, unit packages, intermediate boxes, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129.

5.4.1 Polyethylene bagged packages. Polyethylene bagged packages shall have the required information legibly printed or stamped in black on the bag across the center face or on a white paper label inserted within the bag as to permit ready identification. The bag or label shall indicate the following:

STOCK NUMBER
NOMENCLATURE
QUANTITY

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The flags and pennants covered by this specification are intended to be used by individuals and military organizations for identification or signaling.

6.2 Acquisition requirements.

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. When a first article is required (see 3.2).

d. Title of flag or pennant required.

e. Applicable drawing.



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f. Type of heading required (see 3.5.2, 6.6 and applicable drawing).

g. When a cord and tassel are required (see 3.3.12 and applicable drawing).

h. When a fringe is required (see 3.3.10 and applicable drawing).

i. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

j. Cable number if different than cited on the flag drawing.

k. Type of fabric required if other then cited on flag drawing.

l. Type of distance line required (see 3.3.5 and applicable drawing).

6.3 Napthalene flake requirements. The napthalene flake requirement (see 5.1.1) is not applicable when Government furnished material of wool or part wool content has been moth treated.

6.4 Samples. For access to guide samples, address the contracting activity issuing the invitation for bids.

6.5 Screen printing. With screen printed applications, experience has shown the type of colorants which provide a degree of colorfastness equal to the basic shades of the flag or pennant are derived from the following classes:

(a) Cellulosic fabrics; vat dyed in all cases except for U.S. Army scarlet, U.S. Army maroon and other reds which can be matched only with napthol dyes.

(b) Cellulose reactive dyes and resin-bonded pigment systems may be used successfully in lieu of vat dyes and napthol dyes.

(c) For nylon and nylon-wool buntings, acid and neutral premetallized types have been used.

6.6 Headings. Headings covered by this specification are as shown on the drawings and/or figures below:

| Heading | Drawing | Figure |
|---|---|---|
| Type I | | |
| Class 1 | 5-1-45A | Figure 1 |
| Class 1A | 5-1-45H | Figure 2 |
| Class 2 | 5-1-45B | Figure 3 |
| Class 3 | 5-1-45G | Figure 4 |
| Class 4 | 5-1-45D | Figure 5 |
| Type II | 5-1-45E | Figure 6 |
| Type III | | |
| Class 1 | 5-1-45F | Figure 7 |
| Class 2 | 5-1-45F | Figure 7 |

6.7 First article. When first article inspection is required, the contracting



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officer should provide specific guidance to offerors whether the item should be preproduction sample, a first production sample or a standard production item from the contractor's current inventory as specified in 4.4. The first article should consist of one completed flag or pennant. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. The invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or tests, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.8 Subject term (key word) listing.

Embroidered
Bonnaz
Screen printed
Applique
Distance line
Heading

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - IH
Air Force - 11
Navy - NU

Review activities:
Air Force - 82
DLA - CT

User activity:
Air Force - 45

Preparing activity:
Army - IH

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TIOH DRAWING 5-1-45A FLAG HEADINGS - TYPE I CLASS 1
With Distance Line (Except for Acrylic Flags)

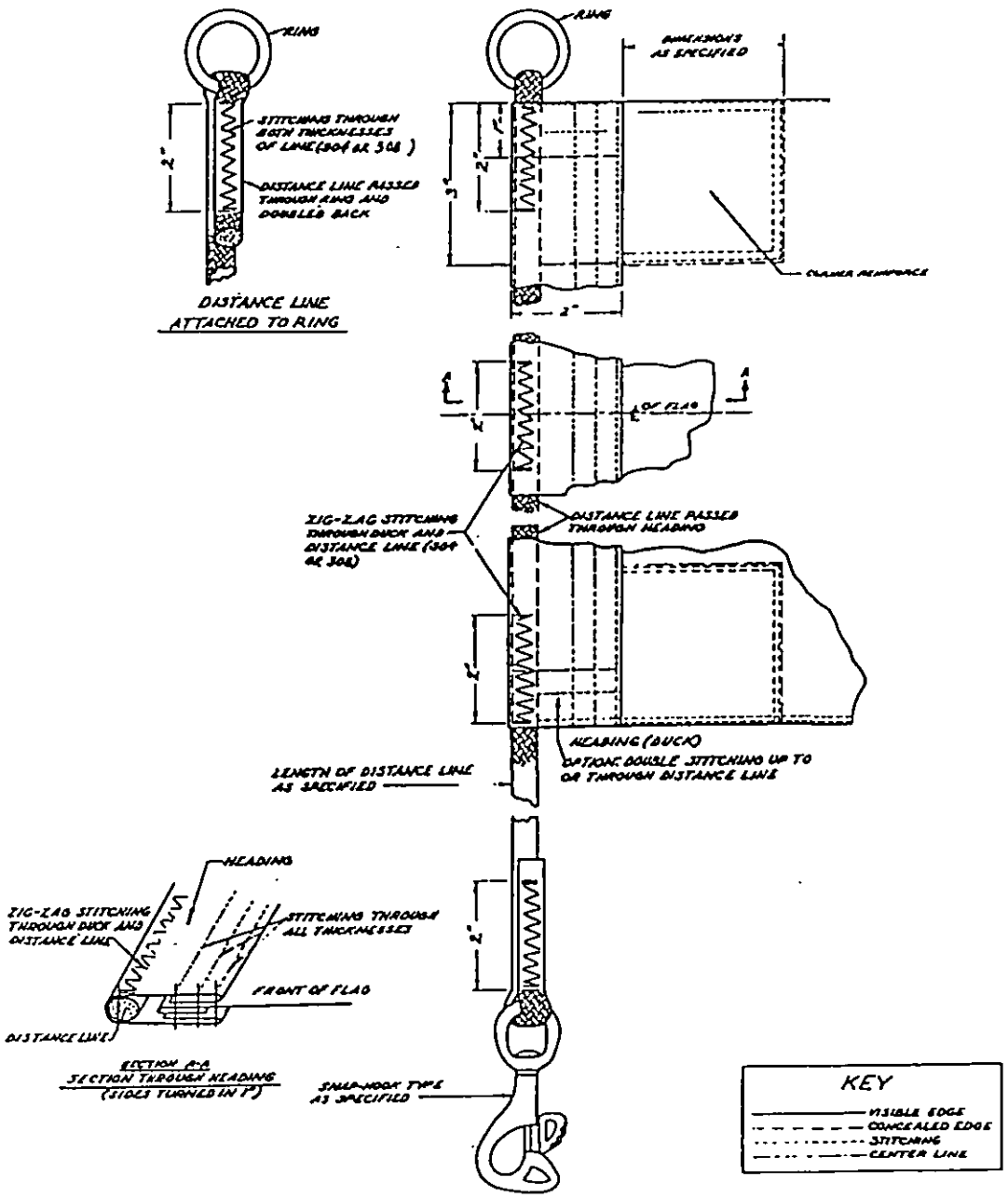



FIGURE 1



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TIOH DRAWING 5-1-45H FLAG HEADINGS - TYPE I CLASS 1A
With Distance Line (For Acrylic Flags)

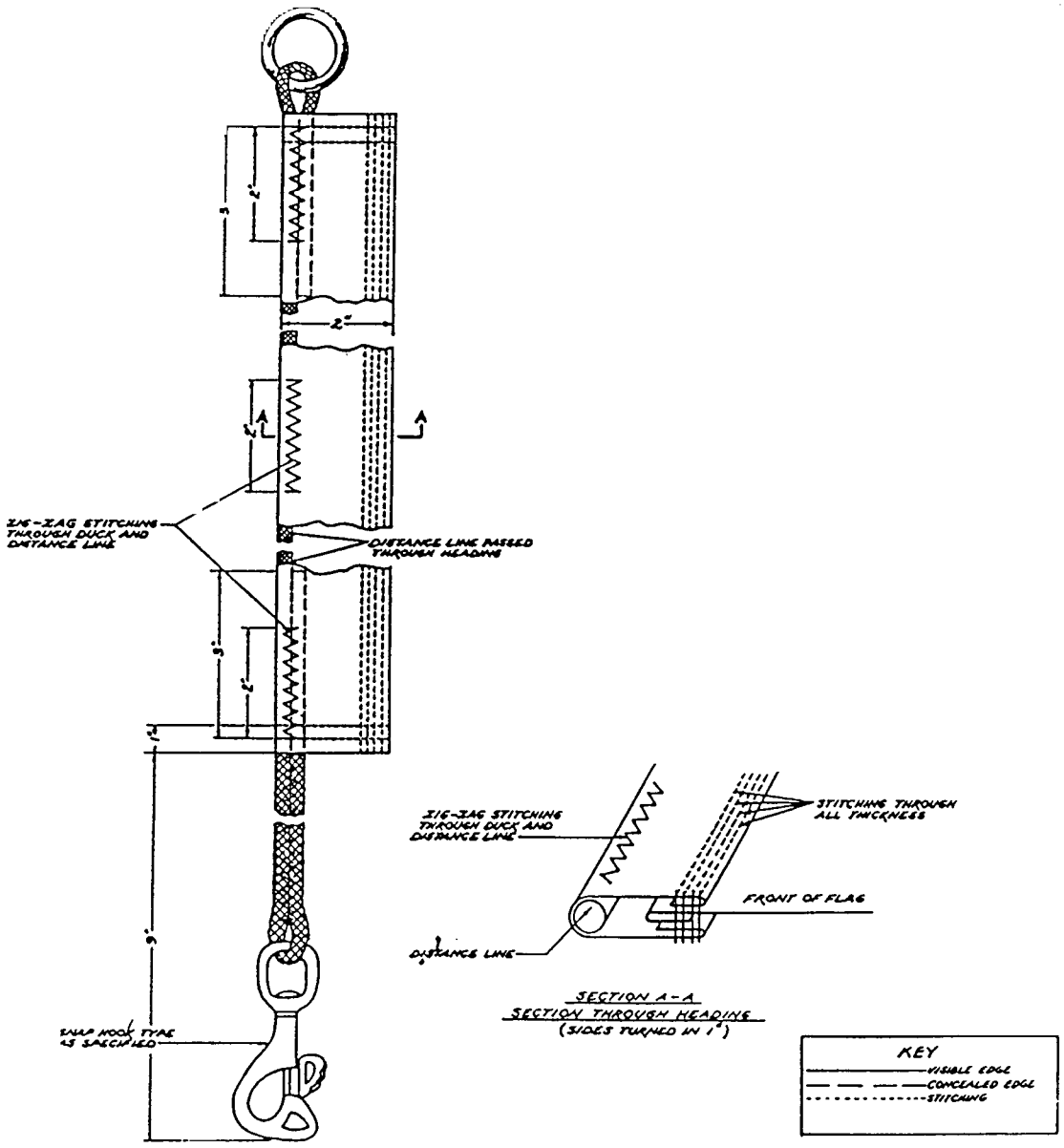



FIGURE 2



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TIOH DRAWING 5-1-45B FLAG HEADINGS - TYPE I CLASS 2
With Grommets

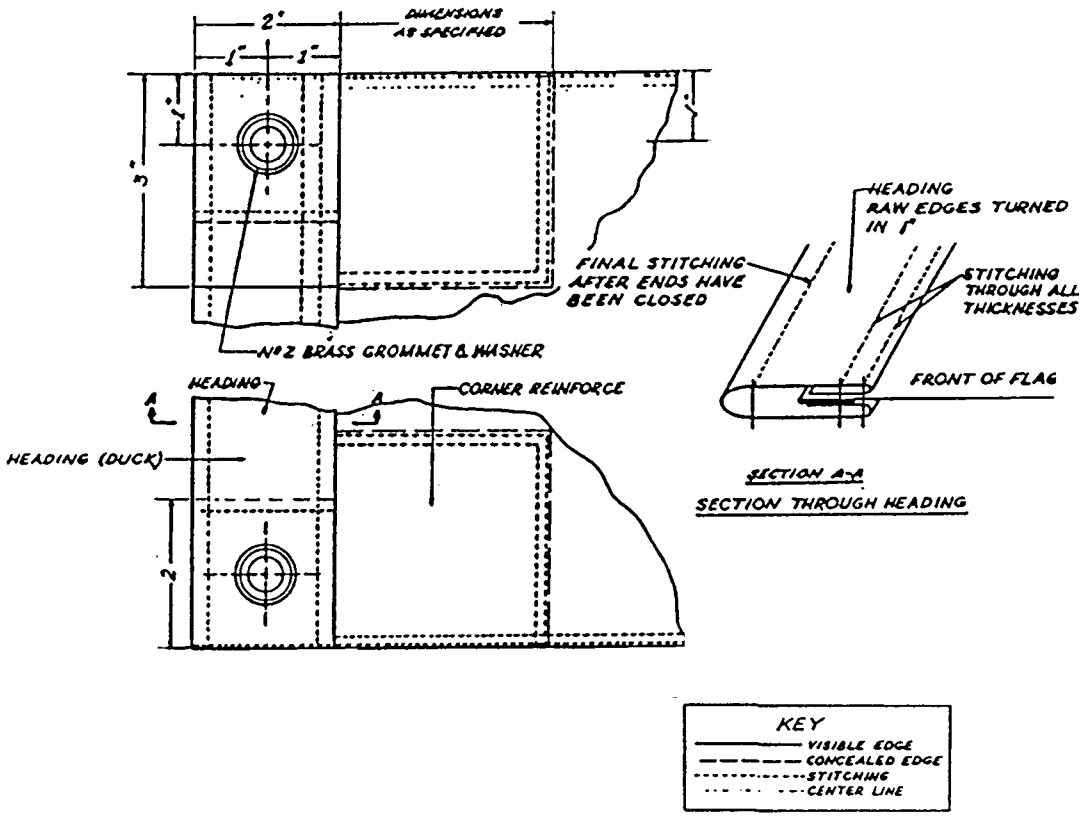
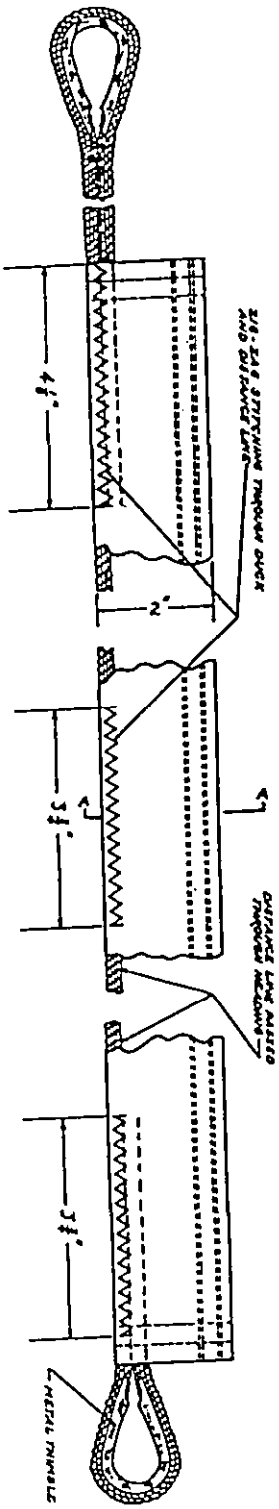


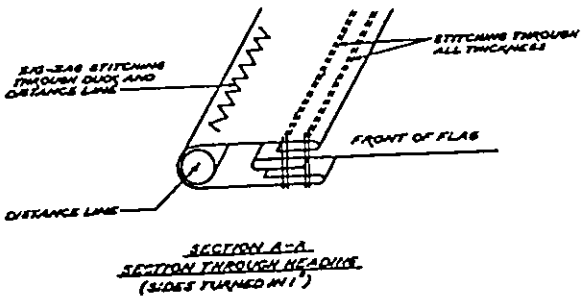

FIGURE 3



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TIOH DRAWING 5-1-45G FLAG HEADINGS - TYPE I CLASS 3
With Distance Lines and Thimbles



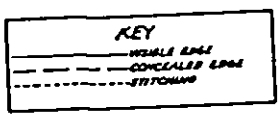




FIGURE 4



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TIOH DRAWING 5-1-45D FLAG HEADINGS - TYPE I CLASS 4
With Grommets and Leather Tabs

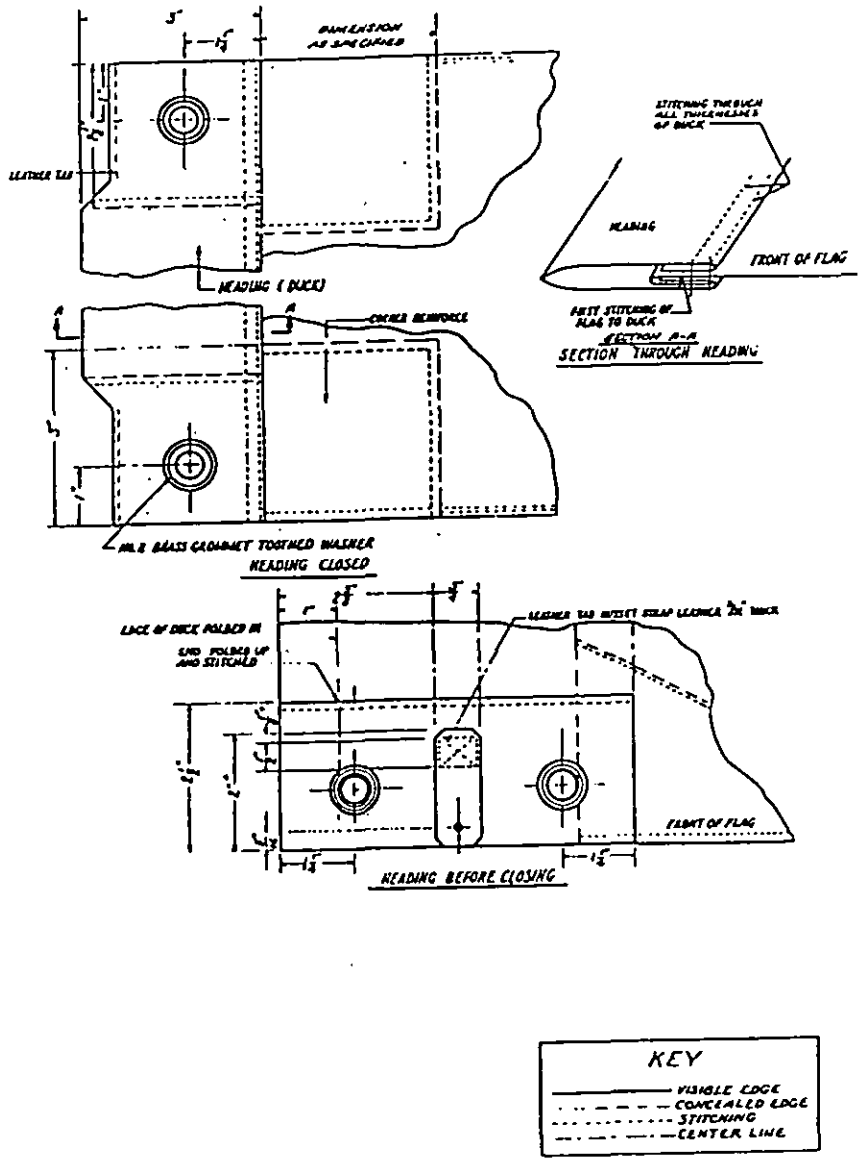



FIGURE 5



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TIOH DRAWING 5-1-45E FLAG HEADINGS - TYPE I
Tube and Tunnel Type (With Cotton Sheeting Lining and Leather Tab)

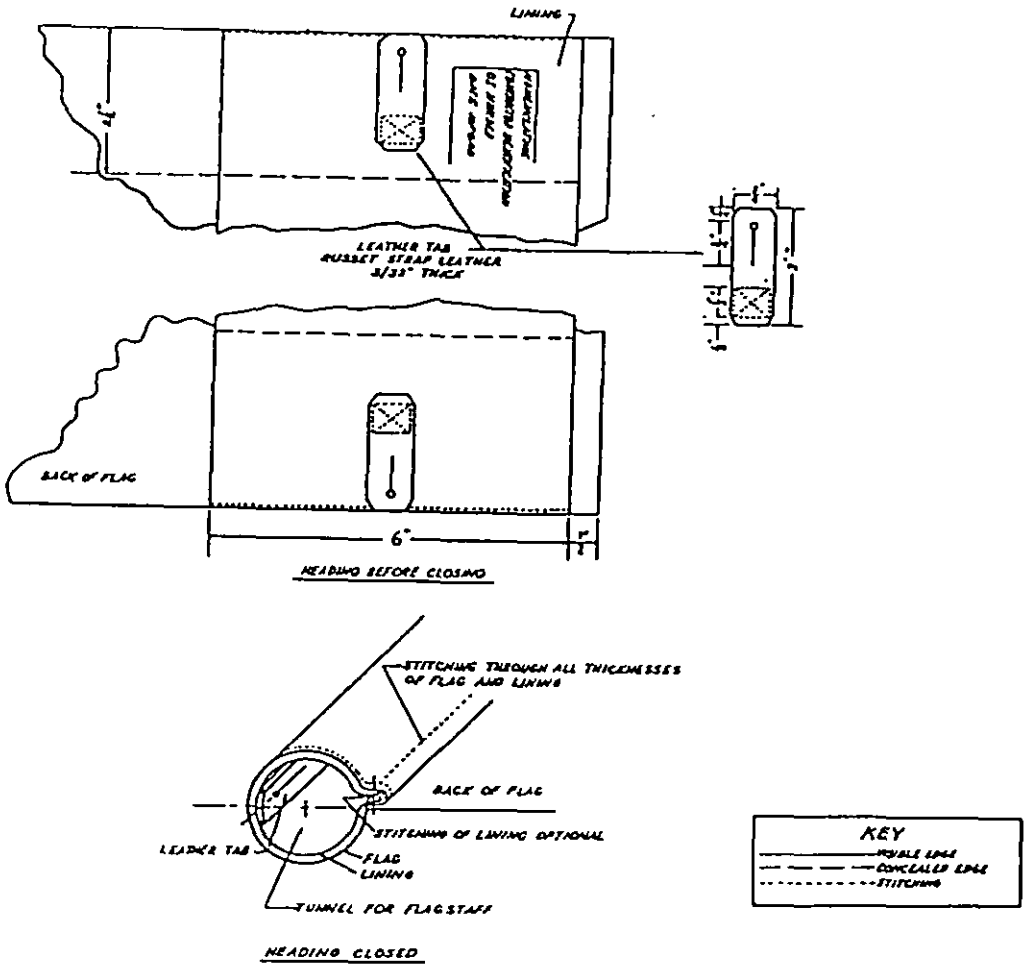


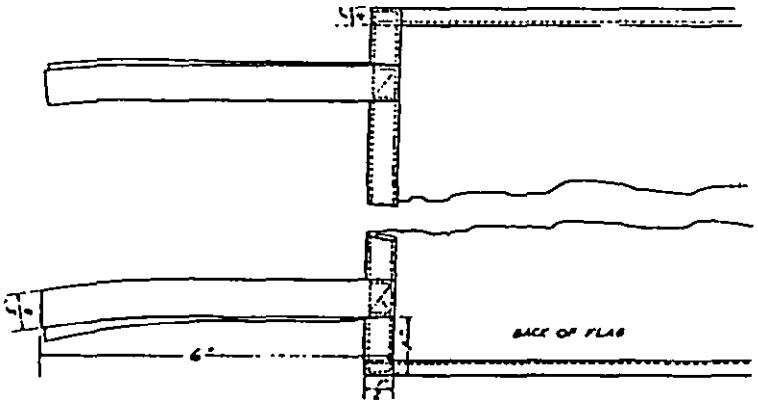

FIGURE 6



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TIOH DRAWING 5-1-45F FLAG HEADINGS - TYPE III CLASS 1 AND CLASS 2

TYPE III CLASS 1 - With Tapes




TYPE III CLASS 2 - With Leather Tabs

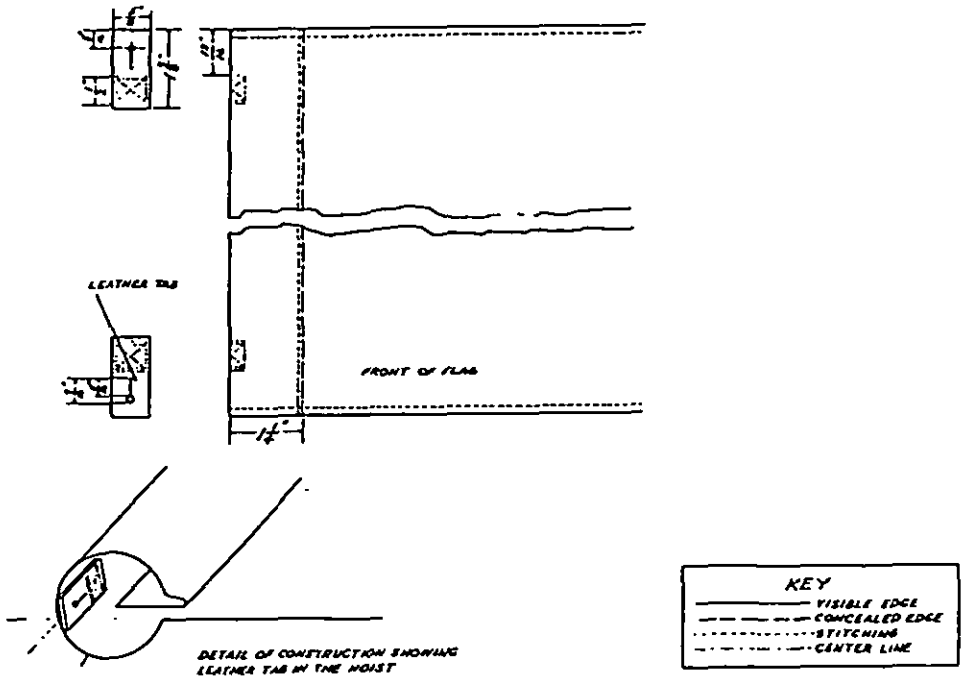
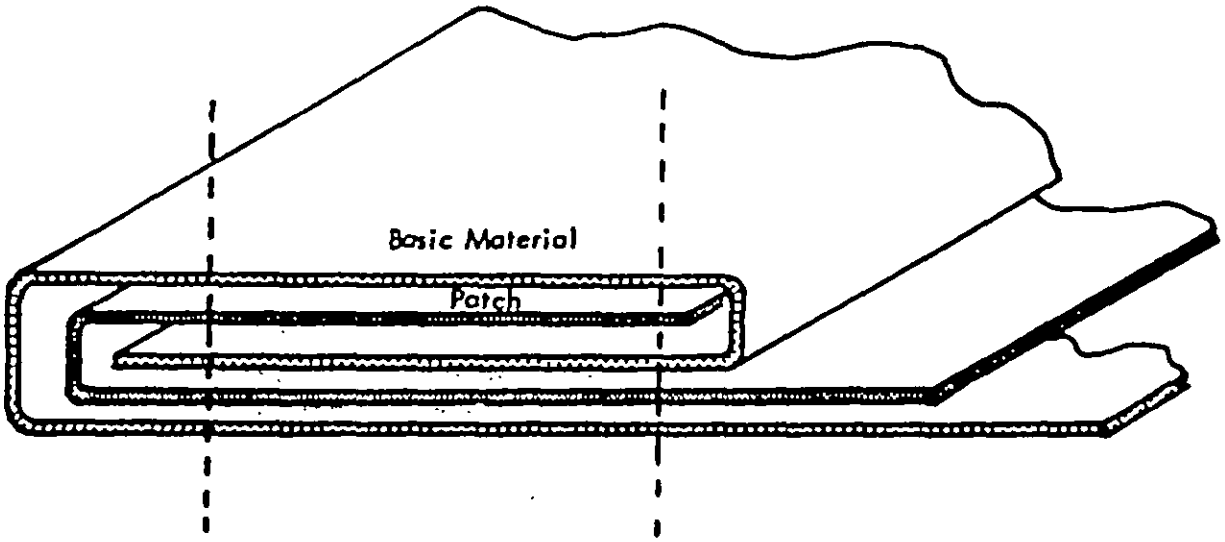



FIGURE 7



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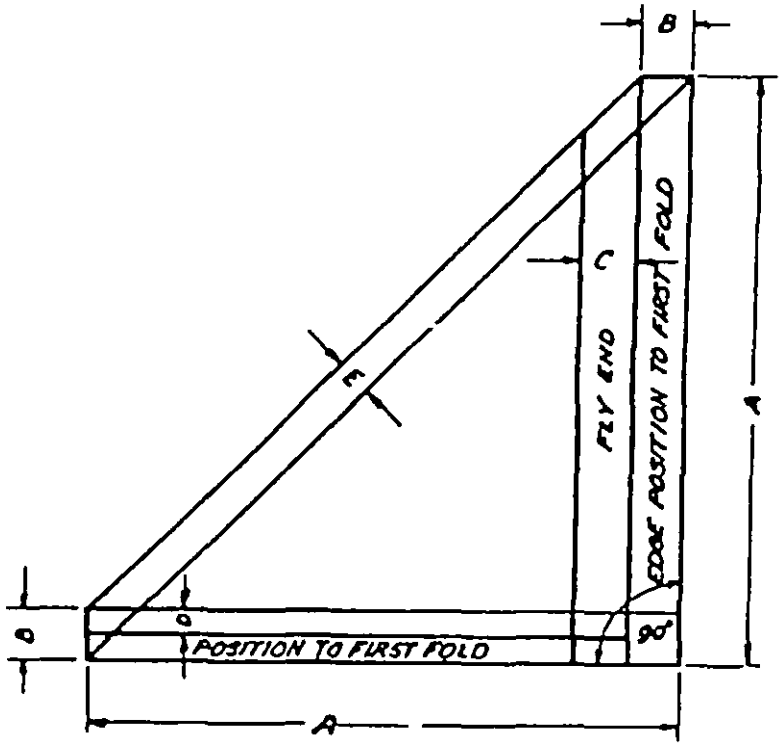



SEWING OF REINFORCEMENT PATCHES TO THE FLY CORNERS OF FLAG

FIGURE 8



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REINFORCEMENT PATCH

| FLAG SIZE | DIMENSIONS IN INCHES | | | | |
|---|---|---|---|---|---|
| | A | B | C | D | E |
| 3½ | 10¼ | ¾ | ¾ | ⅜ | ½ |
| 4 | 9¼ | ¾ | ¾ | ⅜ | ½ |
| 6 | 8¼ | ¾ | ¾ | ⅜ | ½ |

FIGURE 9



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|
| | MIL-F-2692H | 920326 |

3. DOCUMENT TITLE
FLAGS AND PENNANTS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

| 6. SUBMITTER | | |
|---|---|---|
| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON<br>(If applicable) | 7. DATE SUBMITTED (YYMMDD) |

| 8. PREPARING ACTIVITY | | |
|---|---|---|
| a. NAME<br>The Institute of Heraldry | b. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(703) 274-6636 | (2) AUTOVON<br>284-6636 |
| c. ADDRESS (Include Zip Code)<br>Bldg 15, Technical & Production Division<br>Cameron Station<br>Alexandria, VA 22304-5050 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 | |



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